

HARSHITH NARASIMHAMURTHY

Data Scientist · ML Engineer · Data Analyst

harshithnchandan@gmail.com · +91 9663918804 · Bengaluru, India · [LinkedIn](#) · [GitHub](#) · [Webpage](#)

PROFESSIONAL SUMMARY

Data Scientist with 3+ years shipping production ML systems across banking, insurance, and agribusiness. Delivered 5 end-to-end ML products at SatSure including a **geospatial credit scoring model at ~90% accuracy** and an **insurance recovery pipeline driving ~20% uplift** in on-time loan collections. Hands-on across the full **ML lifecycle**: feature engineering, model training, **MLflow** experiment tracking, **FastAPI** deployment, **Docker** containerization, and CI/CD via **GitHub Actions** on AWS. Expanding into **Generative AI with LLMs**, **RAG pipelines**, and **LangChain agentic workflows**. **Gold Medalist, M.Sc. Ag Statistics**.

PROFESSIONAL EXPERIENCE

Data Analyst | [SatSure Analytics](#) · Bengaluru, India | Jul 2023 – Present ·

- ▶ **Delivered 5 production ML systems** across banking, insurance & agribusiness—spanning geospatial credit scoring, crop risk assessment, and insurance recovery analytics; recognized as **Customer Champion** for sustained enterprise impact
- ▶ **Engineered geospatial credit scoring model** (Random Forest + satellite NDVI features) achieving **~90% accuracy** in farmer creditworthiness; adopted by partner banks to accelerate agricultural loan underwriting
- ▶ **Architected semi-automated insurance recovery intelligence pipeline** integrating field-agent workflows with predictive risk scores, driving **~20% uplift in on-time loan recovery revenue** (FY 2025–26)
- ▶ **Built and operationalized scalable ETL/ELT pipelines** on Apache Airflow + AWS (S3, Athena, EC2), cutting data processing latency by **~25%** and improving downstream analytics SLA reliability

ML Research Intern | [Western Sydney University](#) · Sydney, Australia | Jan 2023 – Mar 2023 ·

- ▶ **Developed LSTM & ARIMA time-series forecasting models** for crop yield prediction; engineered preprocessing and cross-validation workflows that improved forecasting accuracy by **~12%** vs. baseline, reducing manual analytical overhead by 40%

KEY ML PROJECTS

Customer Segmentation MLOps Pipeline — End-to-End ML System

- ▶ Architected a full MLOps lifecycle pipeline for customer segmentation using **KMeans clustering** on RFM + behavioral features; tracked experiments with **MLflow + DagsHub**, enabling reproducible model comparison across runs
- ▶ Deployed model via **FastAPI REST API** and Streamlit interactive UI; containerized the full stack with **Docker** and automated build + push to Docker Hub via **GitHub Actions CI/CD**
- ▶ Engineered modular ingestion, validation, transformation, and training pipeline from **MongoDB** source with pytest coverage; live at Streamlit Cloud

Formula 1 AI SQL Assistant — LLM-Powered Text-to-SQL Engine

- ▶ Engineered a production Text-to-SQL agent using **Groq LLaMA 3.3 70B + LangChain** with conversation memory, self-correction logic, and SELECT-only guardrails; enables zero-SQL natural language querying over live structured F1 race data via API

Real-Time Crop Health Monitoring — Production Satellite ML System

- ▶ Deployed a 10m×10m resolution NDVI time-series platform with K-Means cluster-based health classification (Good/Average/Poor); reached **~95% accuracy in damage detection**, adopted by banks and insurers for loan underwriting and yield assurance at scale

Geospatial Bank Branch Expansion — Location Intelligence & ML

- ▶ Built a geospatial ML pipeline (K-Means segmentation + hyperparameter-tuned Random Forest scoring) on large-scale demographic datasets; achieved **~82% location selection accuracy** and ~35% reduction in site evaluation costs, directly informing multi-market branch rollout strategy

TECHNICAL SKILLS

Languages & Query: Python (Pandas, NumPy, Scikit-learn), SQL (advanced analytics, optimization), Prompt Engineering

ML / AI: Predictive Modeling, Feature Engineering, Random Forest, LSTM, ARIMA, Clustering, A/B Testing, Hypothesis Testing, LLMs, RAG, LangChain, Agentic AI

MLOps & Cloud: Apache Airflow, AWS (S3, Athena, EC2, ECR, EKS), ETL/ELT Pipelines, Model Deployment, Apache Iceberg

Visualization & BI: Tableau, Matplotlib, Seaborn

Automation & Dev: N8n Automation, Cursor AI, REST APIs, Git

EDUCATION & RECOGNITION

M.Sc. Agricultural Statistics · Anand Agricultural University · 2021–2023 · **Gold Medalist (University Rank #1)**

B.Sc. Agriculture · University of Agricultural Sciences, Bengaluru · 2017–2021 · **AIR #6**

Certifications: Generative AI with LangChain & HuggingFace · Advanced Python for Data Science · HackerRank SQL · Google Project Management

Publications: Memory-based neural network for cumin price forecasting (Gujarat) · RNN architecture for banana price forecasting (Gujarat)